

Systemic Gender Inequities in Who Reviews Code

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The rapid integration of neurodiversity considerations into software engineering research has highlighted the need to better understand how diverse cognitive profiles shape developer experiences and productivity. Among these, Attention Deficit Hyperactivity Disorder (ADHD)—characterized by differences in attention, working memory, and cognitive processing—affects a significant portion of the developer population, with over 10% of programmers reporting related conditions. While ADHD is associated with strengths such as creativity and hyperfocus, it also introduces challenges in core software development tasks, including time management, abstraction, and sustained attention. Existing workplace tools and practices are often designed with neurotypical assumptions, leading to environments that may inadvertently hinder the effectiveness and well-being of ADHD developers.

Despite growing awareness, there remains a limited and fragmented understanding of how ADHD specifically impacts software development workflows and how developers adapt through personal strategies or organizational support. To address this gap, the paper presents a mixed-methods investigation combining large-scale qualitative analysis of community discussions with a broad survey of professional programmers. This approach enables a deeper examination of the relationships between workplace challenges, coping strategies, and accommodations, as well as their generalizability across developer populations. By systematically mapping these elements, the study aims to inform the design of more inclusive tools, policies, and work practices that not only support ADHD developers but also enhance productivity and collaboration across the software engineering workforce.

1 Motivation

The motivation for this paper arises from the growing recognition that **neurodiversity is both prevalent and under-supported in software engineering**, particularly in the case of ADHD. A substantial portion of developers—over 10%—report attention and memory-related conditions, yet **existing software development environments, tools, and workflows are largely designed with neurotypical assumptions**. This mismatch creates systemic barriers for ADHD developers, where everyday practices such as meetings, open office environments, and notification-heavy workflows can disrupt focus, reduce productivity, and negatively affect well-being. Importantly, these challenges are not merely individual limitations but are shaped by the surrounding work environment, aligning with the broader view that disability is influenced by social and organizational contexts rather than inherent deficits alone.

Another key motivation is the **risk of lost potential due to inadequate support**. ADHD is associated with strengths such as creativity, hyperfocus, and cognitive dynamism, which can be highly valuable in software development. However, without proper accommodations or supportive practices, these strengths often remain underutilized, leading to negative outcomes such as reduced performance, job instability, or disengagement. Moreover, developers are often reluctant to request accommodations due to concerns around stigma, privacy, or workplace bias, meaning that even available support mechanisms are underused. This creates a gap where both individuals and organizations miss out on potential productivity gains that could arise from better inclusion.

The paper is further motivated by a **lack of systematic, task-specific understanding of ADHD in software engineering**. While prior research has explored neurodiversity broadly, it often either aggregates different neurodivergent conditions or does not examine how challenges manifest across specific software tasks such as debugging, design, or code review. Additionally, there is limited knowledge

about how personal coping strategies and organizational accommodations interact with these challenges. Without this mapping, it is difficult for companies to design targeted interventions or for researchers to build effective developer tools tailored to diverse cognitive needs.

Finally, the study is motivated by the broader goal of **improving both inclusivity and overall productivity in software teams**. Understanding how ADHD developers approach problem-solving and adapt to challenges can inform the design of tools and policies that benefit not just neurodivergent developers but the entire workforce. Similar to frameworks like GenderMag, which have successfully improved inclusivity in software design, the authors aim to generate actionable insights that can guide organizational practices and tool development. In doing so, the paper positions neurodiversity not as a constraint but as an opportunity to rethink software engineering practices in a way that is both more inclusive and more effective.

2 Methodology

2.1 Research Design and Overview

The study adopts a **two-phase mixed methods approach** to comprehensively investigate the experiences of ADHD programmers. The first phase involves a **qualitative analysis of community-generated data** to uncover nuanced, real-world experiences, while the second phase uses a **large-scale quantitative survey** to validate and generalize these findings across a broader population of developers. This design enables both depth and breadth, allowing the authors to not only identify key challenges, strategies, and accommodations but also assess their prevalence and significance across different neurotypes.

2.1.1 Phase 1: Qualitative Analysis of Archival Data

In the first phase, the authors conducted an in-depth qualitative study using data from the subReddit rADHD_Programmers, the largest public online community dedicated to ADHD developers. They collected **2,037 posts using the Pushshift API**, of which **881 were identified as software job-related** through manual annotation based on criteria such as references to employment, software tasks, or workplace experiences. From this dataset, **99 posts and their 1,559 associated comments** were selected using a stratified sampling approach to ensure representation across categories like challenges, strategies, accommodations, and disclosure. This resulted in a rich dataset of approximately **160,000 words** for analysis.

The analysis followed a **three-pass thematic coding process**. First, relevant text segments were identified using semantic unitization. Second, high-level codes were applied to categorize the data. Third, detailed coding was performed to extract deeper themes and relationships. The authors developed a comprehensive **codebook consisting of 168 lower-level codes grouped into 11 top-level categories**, including work challenges, coping mechanisms, strengths, and accommodations. To ensure reliability, multiple researchers independently coded the data and resolved discrepancies through negotiated agreement, achieving a **high inter-rater reliability (Krippendorff's $\alpha = 0.75$)**. Additionally, **axial coding** was used to map relationships between challenges, strategies, and accommodations, forming a structured representation of how developers address workplace difficulties.

2.1.2 Phase 2: Quantitative Survey and Validation

To validate and generalize the qualitative findings, the authors designed a **20-minute online survey** targeting professional programmers across different neurotypes. The survey included **Likert-scale questions derived from qualitative themes**, focusing on work challenges, coping strategies, accommodations, and strengths. It was intentionally designed to be inclusive of both ADHD and non-ADHD developers to enable comparative analysis.

Participants were primarily recruited using **public GitHub email addresses**, identifying contributors from top repositories across multiple programming languages. This approach yielded **493 valid responses** after filtering for consistency and quality. The sample included **239 ADHD developers, 60**

other neurodivergent individuals, and 194 neurotypical developers, ensuring a diverse and representative population. The survey also incorporated measures of positive mental health and professional experience to account for potential confounding factors.

2.2 Statistical Analysis and Modeling

The quantitative data was analyzed using a combination of statistical techniques to identify significant differences and relationships. The authors employed the **chi-square test of independence** to compare categorical variables across neurotypes and used **ordinal logistic regression models** for Likert-scale responses, treating them as ordered data. Predictor variables included neurotype, years of experience, mental health, work environment, and task competency. To ensure robustness, the models were tested for assumptions such as proportional odds, and a **significance threshold of $p \leq 0.05$** was applied with **Benjamini-Hochberg correction** to account for multiple comparisons.

2.3 Integration of Qualitative and Quantitative Findings

Finally, the study integrates insights from both phases to produce a **validated mapping between workplace challenges, personal strategies, and organizational accommodations**. The qualitative phase provides rich, contextual understanding, while the quantitative phase confirms the prevalence and generalizability of these findings. This triangulation strengthens the validity of the results and enables the authors to draw broader conclusions about how ADHD impacts software development and how these challenges can be effectively addressed across diverse developer populations.

3 Key Findings

The study reveals that **ADHD programmers experience workplace challenges more frequently and more intensely than neurotypical developers**, with statistically significant differences across all major categories of software development tasks. Specifically, ADHD developers were found to be **1.8 to 4.4 times more likely to struggle with key challenges**, particularly in areas such as time management, design, and sustained attention. Despite this, the findings also highlight that many of these challenges are not exclusive to ADHD developers; rather, they occur across all developer groups, suggesting that addressing them could yield broader productivity benefits. The study identifies five major categories of challenges—cognitive, time-related, distraction-related, communication-related, and alignment-related—each associated with specific software tasks and mitigated through a combination of personal strategies and organizational accommodations.

3.1 Challenges Involving Cognition

A central finding is that **ADHD developers face significant difficulties with memory and abstraction**, which are critical for tasks such as software design, code review, and meetings. These challenges often manifest as difficulty retaining information from discussions or becoming overwhelmed by complex systems. ADHD developers reported struggling more frequently with memory during meetings, design, and documentation, and with abstraction during tasks like debugging and code review. These issues can hinder the ability to form clear mental models, which are essential for effective software development. To cope, developers commonly use strategies such as **task chunking, externalizing information through notes, and visualizing workflows**, while organizations can support them through accommodations like providing additional notes or adjusted training materials.

3.2 Challenges Involving Time

Another major finding is that **time management and task initiation (inertia) are disproportionately challenging for ADHD developers**. These challenges are particularly evident in tasks such as design, documentation, and debugging, where developers may either hyperfocus and lose track of time or struggle to begin or complete tasks. The study shows that ADHD developers are significantly more likely to report issues with managing time and maintaining consistent performance. Common coping strategies

include **using deadlines as motivation, working in structured intervals (e.g., Pomodoro), taking breaks, and collaborating with others (“body doubling”)**. Organizational accommodations such as flexible work hours, adjusted deadlines, and job coaching were found to be effective in mitigating these challenges.

3.3 Challenges Involving Distractions

The findings also emphasize that **ADHD developers are highly susceptible to distractions**, including environmental noise, digital interruptions, and internal thoughts. These distractions are particularly disruptive during meetings, documentation tasks, and debugging, where sustained focus is required. ADHD developers reported significantly higher rates of distraction and difficulty with context switching compared to neurotypical developers. To manage this, developers often **modify their work environment, block distractions using tools like noise-canceling headphones, or introduce controlled stimuli such as music or fidgeting**. Organizational support in the form of adjusted workspaces and assistive technologies further helps mitigate these issues.

3.4 Challenges Involving Communication

Communication challenges were another key finding, with **ADHD developers experiencing more frequent misunderstandings and difficulties in verbal communication**. These issues are most prominent in meetings, interviews, and collaborative tasks, where reliance on verbal instructions can lead to confusion or missed information. ADHD developers often prefer written communication, which provides clarity and allows for better information retention. Strategies such as **increasing communication, asking clarifying questions, and taking control of conversations** were commonly used to address these challenges. Correspondingly, accommodations like structured feedback and alternative communication modes (e.g., written instructions) were identified as beneficial.

3.5 Challenges Involving Alignment

Finally, the study highlights that **misalignment between job roles, tasks, and ADHD work styles significantly impacts motivation and productivity**. Tasks perceived as repetitive, rigid, or lacking novelty—such as documentation or routine meetings—were particularly challenging and often led to disengagement. ADHD developers were also more likely to switch jobs in search of better alignment with their work preferences. However, when aligned, they demonstrated notable strengths, including **creativity, hyperfocus, and strong performance in brainstorming and crisis situations**. This suggests that aligning tasks and environments with individual strengths can unlock significant productivity gains.

3.6 Generalizability and Broader Impact

An important overarching finding is that **many challenges faced by ADHD developers are also experienced by non-ADHD developers, albeit less frequently**. This indicates a “curb-cut effect,” where interventions designed to support ADHD developers—such as flexible schedules, better communication practices, and distraction-reducing tools—can benefit the entire developer population. Additionally, many coping strategies, such as task chunking and taking breaks, were widely used across all groups. This reinforces the idea that improving inclusivity in software engineering environments can lead to **broader improvements in productivity, collaboration, and developer well-being**.

4 Implications

Inclusive Software Engineering Practices A key implication of this study is that **software engineering practices must evolve to better support neurodiverse developers, particularly those with ADHD**. The findings show that many standard workflows—such as synchronous meetings, rigid schedules, and interruption-heavy environments—create unnecessary barriers. As a result, organizations should move toward more inclusive practices, including **asynchronous communication, structured information sharing, and flexible work arrangements**. These changes not only reduce cognitive overload for ADHD developers but also improve clarity and efficiency for teams as a whole. The study

reinforces the idea that inclusivity should be embedded into everyday engineering practices rather than treated as an exception.

Tool and Technology Design The paper highlights important directions for **designing developer tools that actively support focus, memory, and task management**. Since ADHD developers often struggle with distractions, context switching, and information retention, tools can be designed to **externalize cognitive load**, such as through automated meeting summaries, reminders for task resumption, and visualization aids for complex workflows. Additionally, tools that help maintain “flow”—for example, notifying developers when long-running tasks complete—can reduce disruptions caused by waiting times. These implications suggest that **future developer tools should not only optimize performance but also account for cognitive diversity**, enabling developers to work in ways that align with their mental processes.

Organizational Policies and Accommodations The findings indicate that **organizational accommodations play a crucial role in mitigating workplace challenges**, yet they are often underutilized due to stigma or accessibility barriers. This suggests a need for **proactive, universally available support systems** rather than case-by-case accommodations that require disclosure. Examples include flexible work hours, incremental deadlines, access to assistive technologies, and options for different communication modes. By making these supports standard rather than exceptional, organizations can reduce the burden on individuals while fostering a more equitable work environment. Importantly, the study shows that many of these accommodations benefit all developers, not just those with ADHD.

Team Productivity and the “Curb-Cut Effect” Another significant implication is the presence of a “curb-cut effect,” where **interventions designed for ADHD developers improve outcomes for the broader developer population**. Since many challenges—such as distractions, time management, and communication issues—are experienced by all developers, addressing them can lead to **overall productivity gains and better collaboration**. For instance, practices like task chunking, clearer communication, and reduced interruptions were widely adopted across neurotypes. This suggests that investing in inclusive practices is not merely a matter of equity but also a **strategic decision to enhance team performance and efficiency**.

Research and Future Work Finally, the study points to several directions for future research, emphasizing the need for **empirical evaluation of interventions and tools in real-world settings**. While the current work relies on self-reported experiences, further research could measure objective outcomes such as productivity, accuracy, and task completion time. Additionally, the authors suggest using **participatory design approaches** to involve ADHD developers directly in the creation of tools and policies, ensuring that solutions are grounded in real user needs. Expanding research to include diverse populations and contexts will also help refine these insights and support the development of **more inclusive and effective software engineering ecosystems**.